

## UNIT:-2 STRAIGHT LINE

Abhishek Gp Dumka

1. Find the Equation of the straight line passing through the point  $(3, -2)$  and making an angle  $45^\circ$  with the straight line  $6x + 5y = 1$ .
2. Find the angle b/w the lines  $\sqrt{3}x + y = 1$  and  $\sqrt{3}x + y = 1$ .
3. Find the Equation of the straight line passing through the point  $(3, -2)$  perpendicular to the line  $x - 3y + 7 = 0$ .
4. If the points  $(-2, -5)$ ,  $(2, -2)$  and  $(8, k)$  are collinear then the value of  $k$  is equal to.
5. Find the angle between the straight lines  $6x - y = 12$  and  $2x - 8y = 7$ .

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- Find the equation of the line passing through the intersection point of lines  $2x - 3y + 1 = 0$  and  $4x + 7y - 3 = 0$  which makes equal intercepts on the axes.
- If a straight line  $\frac{x}{a} + \frac{y}{b} = 1$  passing through the points  $(8, -9)$  and  $(12, -15)$ , find the value of  $a$  and  $b$ .

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- Find the equation of the straight line which passes through the point  $(4, -3)$  such that the portion of it between the axis is bisected by this point.
- In what ratio is the line through the points  $(2, 3)$  and  $(4, -5)$  divided by the points  $(0, 8)$ ?
- Find the equation of a line passing through point  $(1, 1)$  and perpendicular to the line  $3x + 4y = 12$ .

1. Find the length of the perpendicular drawn from the point  $(-1, 4)$  to the line  $3x + 4y - 5 = 0$ .
2. Find the co-ordinate of points which divide the given segment joining the point  $(x_1, y_1)$  and  $(x_2, y_2)$  internally in the ratio  $m_1 : m_2$ .